

MacOS Sequoia
Overview and Assesment
Written to fulfill the 1st task of
Operating System



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2025

PREFACE

We would like to express our gratitude to God Almighty for the blessings and guidance that have enabled us to complete this academic work titled "**MacOS Sequoia Overview and Assessment.**" This project represents our collective effort as a group and aims to provide insights into MacOS Sequoia, the latest innovation in the Apple ecosystem.

Throughout this learning process, we have gained a deeper understanding of the features and improvements introduced by MacOS Sequoia, as well as how these changes can impact user experience and productivity in the workplace. Specifically, we will highlight various aspects such as user interface, performance, security, and integration with other Apple devices.

This work is inspired by the ever-evolving landscape of technology and the significant role that modern operating systems, like MacOS Sequoia, play in shaping the way we work and interact with technology. We hope that this project can serve as a valuable academic reference for students, developers, and professionals interested in system analysis and its impact on daily routines.

We would like to extend our heartfelt thanks to all those who have supported us during the preparation of this work, including our instructors, peers, and the online community. We acknowledge that this project is still a work in progress, and we recognize the importance of continuous improvement. Constructive feedback from readers is greatly welcomed as we strive to enhance and refine our work in the future.

TABLE OF CONTENTS

PREFACE.....	2
TABLE OF CONTENTS.....	3
TABLE OF FIGURE.....	5
CHAPTER I.....	6
INTRODUCTION.....	6
1.1 Background.....	6
1.2 Problem Domain.....	7
1.3 Writing Methodology.....	7
CHAPTER II.....	8
BASIC THEORY.....	8
2.1 Definition of Operating system.....	8
2.2 What is MacOS.....	9
2.3 Mac OS Types.....	9
CHAPTER III.....	12
PROBLEM ANALYSIS.....	12
3.1 What's New MacOS Sequoia.....	12
• Control your iPhone from your Mac.....	12
• Set up iPhone Mirroring on your Mac.....	12
• Tile windows on Mac.....	13
• Tile a window by pressing the Option key while dragging the window to the highlighted area.....	15
• Tile windows using the green button.....	16
• Tile windows using menu options.....	17
• Use video conferencing features on Mac.....	18
• Share your screen in a video call.....	18
• Use Presenter Overlay.....	19
• Add reactions.....	19

● Use passwords on your Mac.....	22
● Schedule a message on Mac to send later.....	23
● Schedule a message.....	23
● Reschedule a message.....	24
● Edit a scheduled message.....	24
● Delete a scheduled message.....	24
● Solve maths in Notes on Mac.....	25
● Solve expressions.....	26
● Define a variable.....	26
● Solve expressions with variables.....	26
● Choose how Maths Results appear.....	27
CHAPTER IV.....	28
CONCLUSION AND SUGGESTION.....	28
4.1 Conclusion.....	28
4.2 Suggestion.....	28
BIBLIOGRAPHY.....	29
APPENDICES.....	30

TABLE OF FIGURE

Figure 3.1 iPhone Mirroring on Mac.....	10
Figure 3.2 Tile windows on Mac.....	11
Figure 3.3 Tile a window by dragging it to the edge of the screen.....	12
Figure 3.4 Tile windows using Option key.....	13
Figure 3.5 Tile windows using the green button.....	13
Figure 3.6 Tile windows using menu options.....	14
Figure 3.7 Share screen in a video call.....	15
Figure 3.8 Use password on your Mac.....	20
Figure 3.9 Mac send later.....	21
Figure 3.10 Solve in Notes on Mac.....	24
Table 3.1 Add Reaction.....	17, 18, 19

CHAPTER I

INTRODUCTION

1.1 Background

The twenty-first and latest major release of Apple's macOS operating system, the successor to macOS Sonoma. This operating system was announced at WWDC 2024 on June 10, 2024. In line with Apple's tradition of naming macOS releases after California landmarks, this operating system is named after Sequoia National Park, located in the Sierra Nevada mountains(*Wikipedia, 2024*).

Named after California's towering Sequoia trees, was launched on September 16th, 2024, as version 15.0 of Apple's desktop operating system for Mac computers. The operating system is equipped with tools and functionalities that resonate with a broad spectrum of users, from everyday tasks to specialized professional work, especially in the creative domain. At the heart of macOS lies Darwin, an open-source Unix-like operating system. It's powered by the XNU kernel, which oversees memory allocation, process management, disk activities and CPU resource distribution. Above this layer is the Quartz windowing system which manages graphical rendering. The graphical user interface known as Aqua, utilizes the Quartz graphics layer to deliver its distinct visual style and allows users to interact with menus, icons, and windows. Let's take a look at some of the new features(*Kevin, 2024*).

You can use iPhone Mirroring to control your iPhone from your nearby Mac. When iPhone Mirroring is turned on, you can also receive iPhone notifications on your Mac, so you don't miss a thing(*Apple, 2025*).

Apple Intelligence supports a new Writing Tool that helps you find the right words almost anywhere you write. With enhanced language capabilities, you can summarize entire lecture materials in seconds, get short versions of long group threads, and minimize unnecessary distractions with prioritized notifications(*Apple, 2024*).

1.2 Writing Objective

The purpose of this ISAS aims to provide an in-depth understanding of the features and performance of macOS Sequoia, while also analyzing user experience and offering recommendations for upgrade decisions.

1.3 Problem Domain

The point of focus of this paper is to analyse the features of Macintosh Sequoia along with its history for context.

1.4 Writing Methodology

The writing method for this paper uses short journals and literature studies, with facts and opinions gathered primarily from the internet and Apple official documentation to explain its main features and history.

1.5 Writing Framework

Systematics that the author uses in this ISAS investigation, is arranged as follows:

CHAPTER I: INTRODUCTION

Describes the background, objectives, scope, methodology, and writing framework of the paper.

CHAPTER II: BASIC THEORY

Explains OS definition, MacOS definition, and types of MacOS

CHAPTER III: PROBLEM ANALYSIS

Discusses new features in MacOS Sequoia

CHAPTER IV: CONCLUSIONS AND SUGGESTIONS

This chapter contains conclusions that can be drawn from research in problem analysis and contains suggestions also on problem solving.

CHAPTER II

BASIC THEORY

2.1 Definition of Operating system

An operating system is difficult to define, but has been called "the layer of software that manages a computer's resources for its users and their applications". Operating systems include the software that is always running, called a kernel—but can include other software as well. The two other types of programs that can run on a computer are system programs, which are associated with the operating system, but may not be part of the kernel and applications, all other software (*Silberschatz et al., 2021*).

The operating system is the primary software that enables interaction between users and computer hardware. This research aims to examine the meaning, basic functions, development history, and types of operating systems based on a literature review. The evolution of operating systems has taken place through five generations, starting from computers without an operating system to the multi-user and multitasking systems used today. This research also identifies various types of operating systems such as DOS, Windows, and UNIX, as well as the important role of the boot process in ensuring the operating system can run correctly. From the results of this study, it was found that operating systems continue to develop to meet increasingly complex computing needs, such as distributed computing and artificial intelligence (*Ashabul Taufik, 2025*).

2.2 Definition of MacOS

MacOS is the computer operating system (OS) for Apple desktops and laptops. It is a proprietary graphical OS that powers every Mac. OSES interact with a computer's hardware, allocating the resources necessary to complete tasks given to it -- for example, running an application. OSES allocate resources including memory, processing power and file storage (*TechTarget, 2025*).

2.3 Mac OS Types

1. Mac OS X 10.0 – Cheetah (2001)

Mac OS X 10.0 (code Cheetah) is the first major release of macOS, Apple's desktop and server operating system. It was released on March 24, 2001, for a price of \$129 after a public beta. Mac OS X was Apple's successor to the classic Mac OS. It was derived from NeXTSTEP and FreeBSD, and featured a new user interface called Aqua, as well as improved stability and security due to its new Unix foundations. It introduced the Quartz graphics rendering engine for hardware-accelerated animations. Many technologies were ported from the classic Mac OS, including Sherlock and the QuickTime framework. The core components of Mac OS X were open sourced as Darwin.

2. Mac OS X 10.1 – Puma (2001)

Mac OS X 10.1 (code named Puma) is the second major release of macOS, Apple's desktop and server operating system. It superseded Mac OS X 10.0 and preceded Mac OS X Jaguar. Mac OS X 10.1 was released on September 25, 2001, as a free update for Mac OS X 10.0 users. The operating system was handed out for free by Apple employees after Steve Jobs' keynote speech at the Seybold publishing conference in San Francisco. Mac OS X 10.1 was codenamed "Puma" because the internal team thought it was "one fast cat." In January 2002, Apple switched to using Mac OS X as the default OS on all new Macs at the time starting with the 10.1.2 release, replacing Mac OS 9.

3. Mac OS X 10.2 – Jaguar (2002)

Mac OS X Jaguar (version 10.2) is the third major release of macOS, Apple's desktop and server operating system. It superseded Mac OS X 10.1 and preceded Mac OS X Panther. The operating system was released on August 23, 2002. It was available both for single-computer installations and in a "family pack" that allowed five installations on separate computers in one household. Jaguar was the first Mac OS X release to publicly use its code name in marketing and advertisements.

4. Mac OS X 10.3 – Panther (2003)

Mac OS X Panther (version 10.3) is the fourth major release of macOS, Apple's desktop and server operating system. It followed Mac OS X Jaguar and preceded Mac OS X Tiger. It was released on October 24, 2003, with the retail price of US\$129 for a single user and US\$199 for a five user, family license. Panther also included Safari as its default browser, as a change from Internet Explorer in Jaguar.

5. Mac OS X 10.4 – Tiger (2005)

Mac OS X 10.4 Tiger, released on April 29, 2005, was the fifth major version of macOS and successor to Panther, bringing key features like Spotlight, Dashboard, a new Safari, Unified theme, 64-bit support on Power Mac G5, and improved graphics. It sold 2 million copies within 6 weeks and by 2007 was used by 67% of Mac OS X users, making it Apple's most successful OS release at the time. Tiger was the first Mac OS X to support Intel processors and also powered the first Apple TV OS. Replaced by Leopard in October 2007 after 30 months (the longest-running version), its last security update came in 2009. Despite ending support, Tiger remains valued by retrocomputing enthusiasts as the last version supporting Classic Environment (Mac OS 9) and PowerPC G3 processors.

6. Mac OS X 10.5 – Leopard (2007)

Mac OS X Leopard (version 10.5) is the sixth major release of macOS, Apple's desktop and server operating system for Macintosh computers. Leopard was released on October 26, 2007, as the successor of Mac OS X Tiger, and is available in two editions: a desktop version suitable for personal computers, and a server version, Mac OS X Server. It retailed for \$129 for the desktop version and \$499 for Server. Leopard was superseded by Mac OS X Snow Leopard (version 10.6) in 2009. Mac OS X Leopard is the last version of macOS that supports the PowerPC architecture as its predecessor, Mac OS X Snow Leopard, functions solely on Intel based Macs.

7. Mac OS X 10.6 – Snow Leopard (2009)

Mac OS X Snow Leopard (version 10.6) (also referred to as OS X Snow Leopard) is the seventh major release of macOS, Apple's desktop and server operating system for Macintosh computers. Snow Leopard was publicly unveiled on June 8, 2009, at Apple's Worldwide Developers Conference. On August 28, 2009, it was released worldwide, and was made available for purchase from Apple's website and retail stores at the price of \$29 USD for a single-user license. As a result of its low price, initial sales of Snow Leopard were significantly higher than its predecessors, which had prices starting at \$129 USD. The release of Snow Leopard came nearly two years after the launch of Mac OS X Leopard, the second longest time span between successive Mac OS X releases (the time span between Tiger and Leopard was the longest).

8. Mac OS X 10.7 – Lion (2011)

OS X Lion, also known as Mac OS X Lion, (version 10.7) is the eighth major release

of macOS, Apple's desktop and server operating system for Mac computers. A preview of OS X 10.7 Lion was publicly shown at the "Back to the Mac" Apple Special Event on October 20, 2010. It brought many developments made in Apple's iOS, such as an easily navigable display of installed applications, to the Mac, and includes support for the Mac App Store, as introduced in Mac OS X 10.6 Snow Leopard version 10.6.6. On February 24, 2011, the first developer's preview of Lion (11A390) was released to subscribers to the Apple Developer program. Other developer previews were subsequently released, with Lion Preview 4 (11A480b) being released at WWDC 2011.

9. OS X 10.8 – Mountain Lion (2012)

OS X Mountain Lion, released on July 25, 2012, was the ninth major macOS version and a refinement of OS X Lion. It focused on tighter integration with Apple devices and iOS-like features. Key additions included Gatekeeper (malware protection), iCloud and Game Center integration, Safari 6, standalone Notes and Reminders, Messages replacing iChat, Notification Center, and built-in Twitter (later Facebook) sharing. It received positive reviews for speed, Messages, and Notification Center, though iCloud was criticized for unreliability. Mountain Lion sold 3 million copies in four days and 28 million total by 2013, making it Apple's most popular OS X release. It was also the last paid OS X upgrade, later offered for free (since 2021), like its predecessor Lion.

10. OS X 10.9 – Mavericks (2013)

OS X Mavericks, announced on June 10, 2013 and released on October 22, 2013, was the 10th major macOS release and the first to be offered free via the Mac App Store. Named after a surfing spot in California, it focused on better battery life, Finder enhancements, stronger iCloud integration, and new iOS apps on Mac. It was the last version with a skeuomorphic design and the Lucida Grande system font, before Yosemite's redesign. Mavericks also began Apple's tradition of naming macOS after California locations.

11. OS X 10.10 – Yosemite (2014)

OS X Yosemite (*/joo'semiti/ yoh-SEM-it-ee*; version 10.10) is the eleventh major release of macOS, Apple Inc.'s desktop and server operating system for Macintosh computers. OS X Yosemite was announced and released to developers on June 2, 2014, at WWDC 2014 and released to public beta testers on July 24, 2014. Yosemite

was released to consumers on October 16, 2014. Following the Northern California landmark-based naming scheme introduced with OS X Mavericks, Yosemite is named after the national park.

12. OS X 10.11 – El Capitan (2015)

OS X El Capitan, released on September 30, 2015 as a free upgrade, was the 12th major macOS version (and the last branded as “OS X”). Named after the rock formation in Yosemite National Park, it focused on performance, stability, and security, and received much better reviews than Yosemite. First shown at WWDC 2015, it went through several developer and public betas before launch. El Capitan was also the final OS X version supporting aluminum Macs and Xserve, since its successor macOS Sierra dropped compatibility with those models.

13. macOS 10.12 – Sierra (2016)

macOS Sierra (version 10.12) is the thirteenth major release of macOS (formerly known as OS X and Mac OS X), Apple Inc.'s desktop and server operating system for Macintosh computers. The name "macOS" stems from the intention to unify the operating system's name with that of iOS, watchOS and tvOS. Sierra is named after the Sierra Nevada mountain range in California and Nevada. Specifically, Lone Pine Peak is the location for macOS Sierra's default wallpaper. Its major new features concern Continuity, iCloud, and windowing, as well as support for Apple Pay and Siri.

14. macOS 10.13 – High Sierra (2017)

macOS High Sierra (version 10.13) is the fourteenth major release of macOS, Apple Inc.'s desktop operating system for Macintosh computers. macOS High Sierra was announced at the WWDC 2017 on June 5, 2017 and was released on September 25, 2017. The name "High Sierra" refers to the High Sierra region in California. Its name signified its goal to be a refinement of the previous macOS version, macOS Sierra, focused on performance improvements and technical updates rather than features. This makes it similar to previous macOS releases Snow Leopard, Mountain Lion and El Capitan. Among the apps with notable changes are Photos and Safari. macOS High Sierra is the final version of macOS to support the Unibody iMac and the Polycarbonate Unibody MacBook, as its successor, macOS Mojave, drops support for the late 2009 and final models.

15. macOS 10.14 – Mojave (2018)

macOS Mojave, announced on June 4, 2018 at WWDC and released on September 24, 2018, was the 15th major macOS version, named after the Mojave Desert. It introduced Dark Mode, brought several iOS apps (News, Voice Memos, Home) to Mac, and was the last version to include iTunes, Dashboard, 32-bit app support, iPhoto, and HFS+ booting without patches. It succeeded High Sierra, preceded Catalina, and received positive reviews with updates after launch.

16. macOS 10.15 – Catalina (2019)

macOS Catalina (version 10.15) is the sixteenth major release of macOS, Apple Inc.'s desktop operating system for Macintosh computers. It is the successor to macOS Mojave and was announced at WWDC 2019 on June 3, 2019 and released to the public on October 7, 2019. Catalina is the first version of macOS to support only 64-bit applications and the first to include Activation Lock. It is also the last version of macOS to have the major version number be 10.x; its successor, Big Sur, released on November 12, 2020, is version 11. In order to increase web compatibility, Safari, Chromium and Firefox have frozen the OS in the user agent running in subsequent releases of macOS at 10.15.7 Catalina. The operating system is named after Santa Catalina Island, which is located off the coast of southern California. macOS Catalina is the final version of macOS that supports the Unibody MacBook Pro, as its successor, macOS Big Sur, drops support for its mid-2012 and final model.

17. macOS 11 – Big Sur (2020)

macOS Big Sur, announced on June 22, 2020 and released on November 12, 2020, was the 17th major macOS version and the first to shift from version 10 to version 11 after two decades. Named after the Big Sur region in California, it introduced a major UI redesign with new icon styles, visual hierarchy, and the addition of Control Center, along with a revamped Time Machine. It was also the first macOS to support ARM-based Macs, marking Apple's transition from Intel. Big Sur was the last macOS to support Nvidia graphics cards, with Monterey dropping that support.

18. macOS 12 – Monterey (2021)

macOS Monterey (version 12) is the eighteenth major release of macOS, Apple's desktop operating system for Macintosh computers. The successor to macOS Big Sur, it was announced at WWDC 2021 on June 7, 2021, and released on October 25, 2021. macOS Monterey was succeeded by macOS Ventura, which was released on October 24, 2022. The operating system is named after Monterey Bay, continuing the trend of

releases named after California locations since 2013's 10.9 Mavericks. macOS Monterey is the final version of macOS that supports the 2015–2017 MacBook Air, Retina MacBook Pro, 2014 Mac Mini, 2015 iMac and cylindrical Mac Pro, as its successor, macOS Ventura, drops support for those models. It is the last version of macOS that can run on Macs with 4GB of RAM.

19. macOS 13 – Ventura (2022)

macOS Ventura (version 13) is the nineteenth major release of Apple's macOS, announced at WWDC on June 6, 2022, and released on October 24, 2022. it succeeded macOS Monterey and was later followed by macOS Sonoma. named after the city of Ventura in California, it's the last macOS version to support most 2017 Macs (except iMac Pro, which is supported until macOS Sequoia).

20. macOS 14 – Sonoma (2023)

macOS Sonoma (version 14) is the twentieth major release of Apple's macOS, announced at WWDC on June 5, 2023, and released on September 26, 2023. named after California's Sonoma County, it succeeded macOS Ventura and was later followed by macOS Sequoia. Sonoma is the last macOS version to support the 2018–2019 MacBook Air.

21. macOS 15 – Sequoia (2024)

macOS Sequoia, announced at WWDC 2024 on June 10 and released on September 16, 2024, is the 21st major macOS version, named after Sequoia National Park. It was the first macOS released on the same day as iOS and iPadOS updates. Sequoia will be the last version to support several Intel-based Macs (iMac Pro, Mac mini, MacBook Pro 2018–2020, 2019 iMac, and Intel MacBook Air). It will be succeeded by macOS Tahoe in September 2025. ([Wikipedia.org](https://en.wikipedia.org/wiki/MacOS) MacOS, 2024)

2.4 Definition of WWDC

Apple's Worldwide Developers Conference, commonly referred to as WWDC, is one of the most important events in the tech world. Held annually, this event brings together developers, engineers, and technology enthusiasts from around the globe to learn about Apple's latest advancements in software and hardware.

The first WWDC took place in 1983, initially known as the Apple Independent Software Developers Conference. This early iteration was a small gathering focused on software development for Apple products. Attendees were required to sign a confidentiality agreement, and the event was shrouded in secrecy. (Routinehub, 2024)

CHAPTER III

PROBLEM ANALYSIS

3.1 New Features in MacOS Sequoia

- **Control your iPhone from your Mac**

You can use iPhone Mirroring to control your iPhone from your nearby Mac. When iPhone Mirroring is turned on, you can also receive iPhone notifications on your Mac, so you don't miss a thing.



Figure 3.1 iPhone Mirroring on Mac, Source: Apple Support (2024), ref at: <https://support.apple.com/en-gb/guide/mac-help/mchl444d53a6/mac>

Note: Some iPhone features such as the camera and microphone aren't compatible with iPhone Mirroring. Also, some Continuity features like Continuity Camera and AirPlay aren't available on your Mac when using iPhone Mirroring.

Before you begin, make sure of the following:

- Both devices have Wi-Fi and Bluetooth® turned on.
- You're signed in to the same Apple Account on both devices.
- Your devices are within 10 metres (30 feet) of each other.
- Your Mac has macOS 15 Sequoia or later and your iPhone has iOS 18 or later.
- Both devices meet minimum system requirements for iPhone Mirroring.
- Set up iPhone Mirroring on your Mac.

- **Tile windows on Mac**

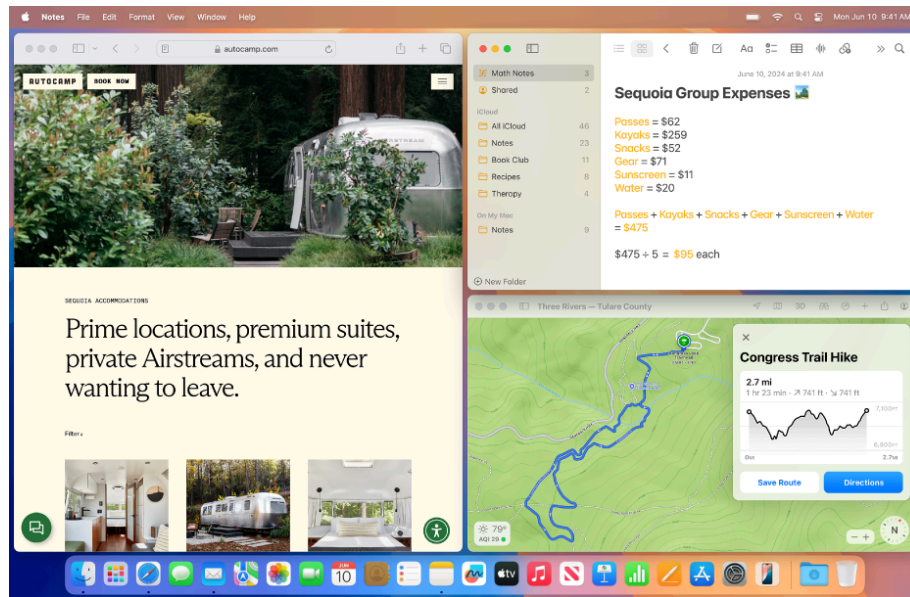


Figure 3.2 Tile windows on Mac, Source: Apple Support (2024), ref at: <https://support.apple.com/en-gb/guide/mac-help/mchlefc287e5d/15.0/mac/15.0>

In macOS Sequoia 15 or later, you can use any of the following methods to tile windows on Mac:

- Drag the window to the edge of the screen.
 - Press the Option key while dragging the window to the highlighted area.
 - Use the green button in the top-left corner of the window.
 - Use menu options.
 - Use keyboard shortcuts. See Mac window tiling icons & keyboard shortcuts.
- **Tile a window by dragging it to the edge of the screen**
You can tile a window by dragging it to an edge of the screen — left, right or any corner, or to the menu bar at the top of the screen. When you drag a window to the menu bar, it fills the desktop.

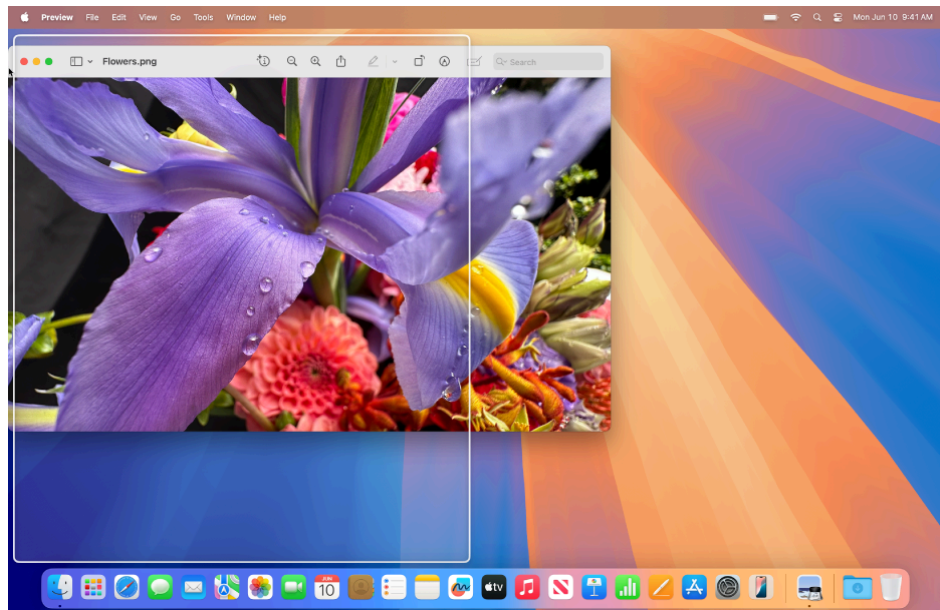


Figure 3.3 Tile a window by dragging it to the edge of the screen, Source: Apple Support (2024), ref at: <https://support.apple.com/en-gb/guide/mac-help/mchlef287e5d/15.0/mac/15.0>

1. Drag a window to an edge of the screen.
2. Release the window when the highlighted area appears.

The window snaps into position when the pointer reaches the edge of the screen.

Note: When you drag a window all the way through the top edge of the screen, Mission Control opens. If you don't want Mission Control to open when you do that, turn off "Drag windows to top of screen to enter Mission Control" in Desktop & Dock settings. See Change Desktop & Dock settings.

You can turn this window tiling feature on or off. See Change window tiling settings.

- **Tile a window by pressing the Option key while dragging the window to the highlighted area**

For faster window tiling, press the Option key while dragging a window towards an edge of the screen — top, left, right — or any corner of the screen.

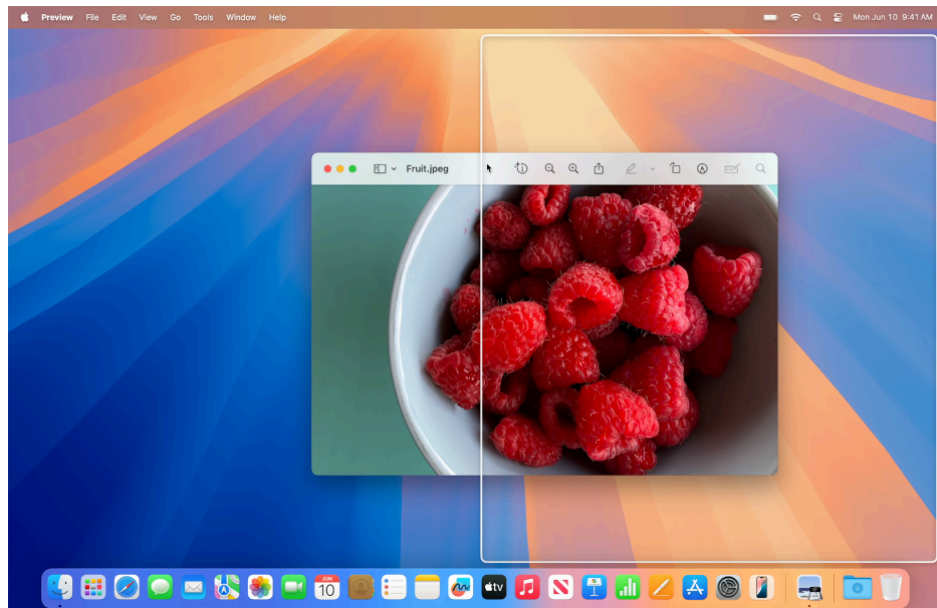


Figure 3.4 Tile windows using Option key, Source: Apple Support (2024), ref at: <https://support.apple.com/en-gb/guide/mac-help/mchlef287e5d/15.0/mac/15.0>

1. Press and hold the Option key.
2. Drag the window towards the area you want to place it in.
3. Release the window when it's in the highlighted area.

You can turn this window tiling feature on or off. See [Change window tiling settings](#).

- **Tile windows using the green button**

The green button in the upper-left corner of each window offers a number of layout options you can use to quickly tile one or more windows.

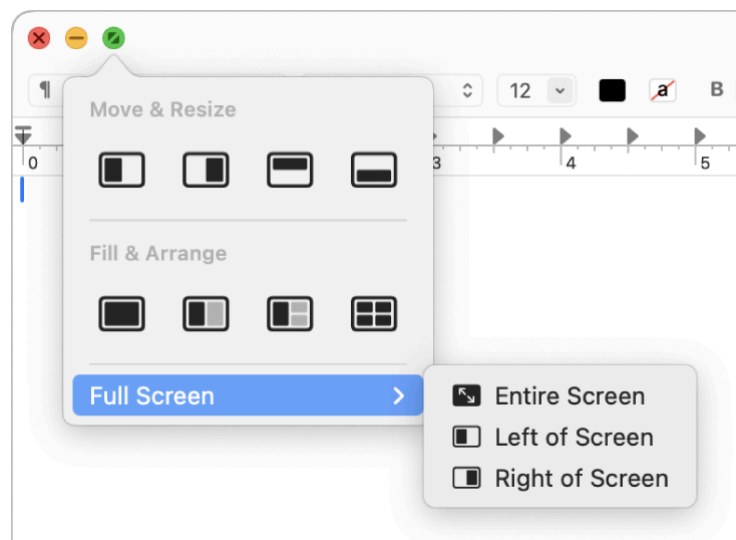


Figure 3.5 Tile windows using the green button, Source: Apple Support (2024), ref at: <https://support.apple.com/en-gb/guide/mac-help/mchlef287e5d/15.0/mac/15.0>

1. Hold the pointer over the green button in the upper-left corner of the active window.
 2. If you want to see additional layouts, press the Option key.
 3. Do one of the following:
 - *Tile the active window:* Click a layout below Move & Resize.
 - *Tile multiple windows:* Click a layout below Fill & Arrange.
 - *Tile full screen:* Hold the pointer over Full Screen, then choose an option.
 - *Centre the active window:* Press the Option key, then click  below Fill & Arrange.
 - *Fill the desktop with the active window:* Click  below Fill & Arrange.
- **Tile windows using menu options**

You can tile one or more windows by choosing options from the Window menu in the menu bar.

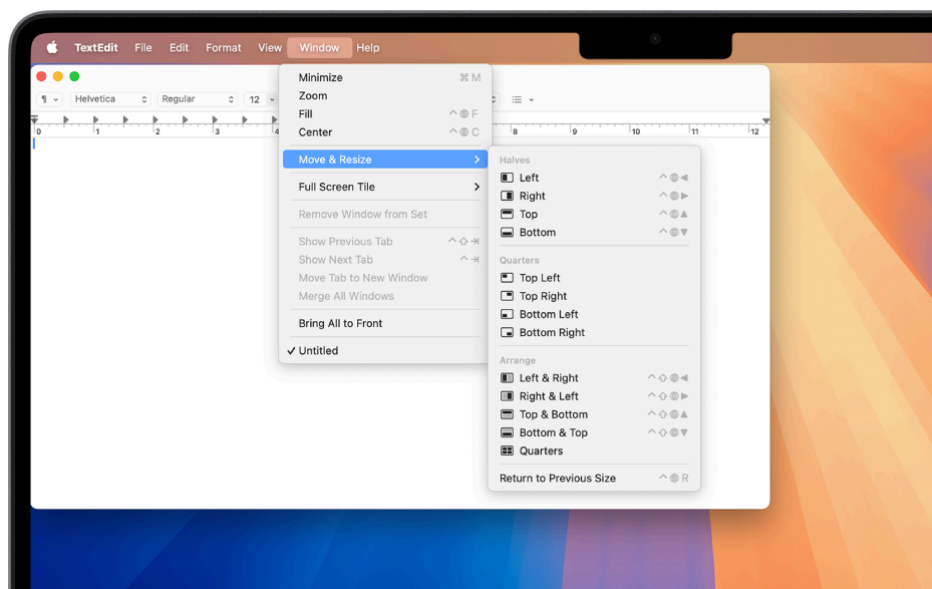


Figure 3.6 Tile windows using menu options, Source: Apple Support (2024), ref at: <https://support.apple.com/en-gb/guide/mac-help/mchlef287e5d/15.0/mac/15.0>

1. Make sure the window you want to tile is active.
2. Choose Window in the menu bar, then do one of the following:

- *Tile the active window:* Hold the pointer over Move & Resize, then choose a layout below Halves or Quarters.
 - *Tile multiple windows:* Hold the pointer over Move & Resize, then choose a layout below Arrange. Press the Option key to see additional layouts below Arrange.
 - *Centre the active window on the desktop:* Choose Centre.
 - *Fill the desktop with the active window:* Choose Fill.
 - *Tile full screen:* Hold the pointer over Full Screen Tile, then choose an option.
- **Use video conferencing features on Mac**

Depending on your Mac model, you may be able to use video conferencing features like screen sharing, Presenter Overlay, video effects and reactions while you use an app that captures video. If you have a supported iPhone, you can also use your iPhone as a webcam on Mac with video conferencing features.

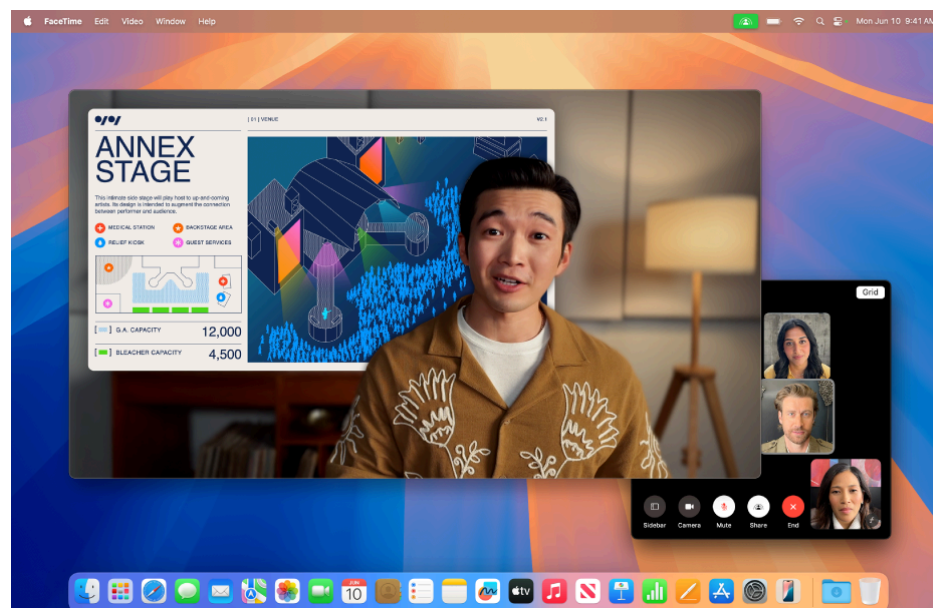


Figure 3.7 Share screen in a video call, Source: Apple Support (2024), ref at: <https://support.apple.com/en-gb/guide/mac-help/mchlff01c919/15.0/mac/15.0>

Share your screen in a video call

1. Go to an app that captures video, such as FaceTime, and make a call.
2. Click  in the menu bar, click , then do any of the following:

- a. *Share the entire screen:* Share your entire screen in the video call. Click Share Entire Screen.
 - b. *Share a window:* Share one app window. Hold the pointer over the window you want to share, then click Share This Window.
 - c. *Share an app:* Share all the windows of a single app. Hold the pointer over a window from the app you want to share, then click Share All Application Windows.
3. When you no longer need to share your screen, click Stop Sharing.

- **Use Presenter Overlay**

If you are using a Mac computer with Apple silicon or Continuity Camera with an iPhone 12 or later as a webcam, you can use Presenter Overlay to customise the appearance of your video while sharing your screen.

1. Go to an app that captures video, such as FaceTime, and make a call.
2. Click  in the menu bar, then click . Select one of the following options under Presenter Overlay:
 - a. *None:* Your video appears in a window that's separate from your shared screen.
 - b. *Small:* Your video appears in a movable bubble that shows a cropped view of you within your shared screen.
 - c. *Large:* Your video is the most prominent feature in the window and your screen is framed next to you on a separate layer.
3. When you've finished with your presentation and no longer need to share your screen, click Stop Sharing.

Note: If you don't see Presenter Overlay as an option, the app you're using doesn't support it.

- **Add reactions**

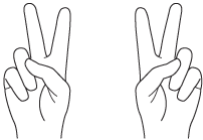
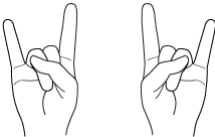
When using an app that captures video, such as FaceTime, you can add reactions that fill the camera frame with fun, 3D augmented reality effects — including hearts, balloons, fireworks and more.

1. Go to an app that captures video, such as FaceTime.
2. Click  in the menu bar, then click  next to Reactions and select a reaction.

If you can't see the arrow, click Reactions to turn on the feature.

3. Click a reaction, or use a hand gesture to generate a reaction. Choose any of the following reactions:

Reaction	Gesture	Icon
Hearts		
Thumbs-up		
Thumbs-down		

Reaction	Gesture	Icon
Balloons		
Confetti		
Laser		
Fireworks		

Reaction	Gesture	Icon
Rain		

Table 3.1 Add Reaction, Source: Apple Support (2024), REF:
<https://support.apple.com/en-gb/guide/mac-help/mchlff01c919/15.0/mac/15.0>

Note: When using gestures, hold your hands away from your face, and pause for a moment to activate the effect.

- **Use passwords on your Mac**

In the Passwords app , you can find all your passwords, passkeys and verification codes in one place. You can use them on all your devices where you sign in to the same Apple Account and turn on Passwords in iCloud settings. And if you use AutoFill to sign in to apps and websites, your passwords automatically appear in Passwords.

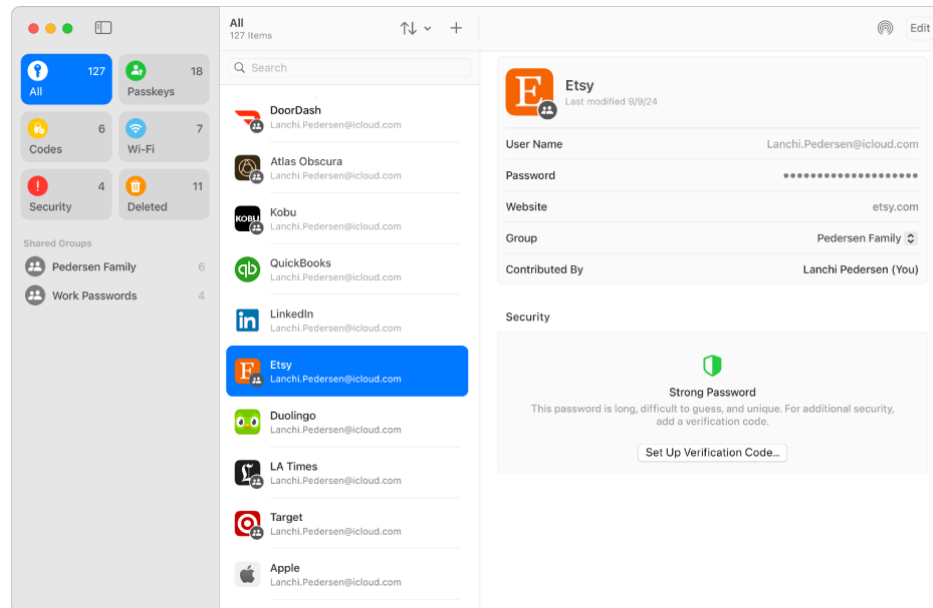


Figure 3.8 Use password on your Mac, Source: Apple Support (2024), ref: <https://support.apple.com/en-gb/guide/passwords/mchl901b1b95/mac>

See all your types of password credentials:

- App passwords
- Website passwords
- Sign in with Apple
- Passkeys
- Verification codes
- Wi-Fi passwords
- Passwords shared with groups

Find a password by sorting or searching. Move the pointer over a password to reveal it. Click Edit to change or delete the password.

Tip: In Passwords, click Security for recommendations to change weak passwords.

- **Schedule a message on Mac to send later**

If it's too late at night or too important to forget, you can use iMessage to schedule a message to be sent later.

Note: You must be online to schedule or modify messages to be sent later, and using iMessage with macOS Sequoia, iOS 18, iPadOS 18, watchOS 11, visionOS 2 or later. Your recipient can use any device, and they won't know that the message was scheduled.

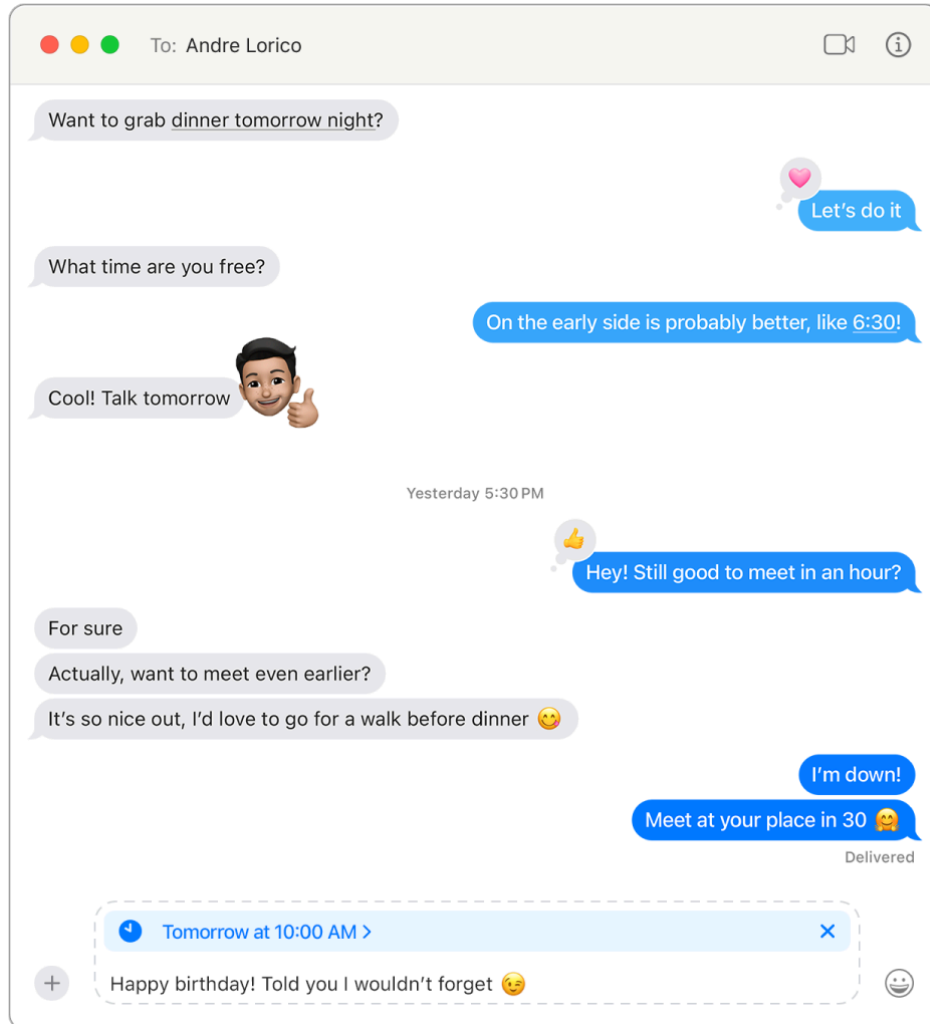


Figure 3.9 Mac send later, Source: Apple Support (2024), ref at: <https://support.apple.com/en-gb/guide/messages/icht113734b8/mac>

- **Schedule a message**

You can schedule a message up to 14 days into the future.

1. Go to the Messages app  on your Mac.
2. Select a conversation.
3. Click , then click Send Later.
4. Choose the date and time you want to send your message.

5. Enter a message, then press Return to schedule the message.

The message appears with a dashed line around it and is stored on a server until it's time to send it — the time appears above the message.

- **Reschedule a message**

You can change when a message is sent until its delivery time.

1. Go to the Messages app  on your Mac.
2. Select the conversation with the message you want to reschedule.
If you can't see your message, scroll down to the bottom of the conversation. Scheduled messages may appear below sent messages.
3. Click Edit in the conversation (above your scheduled message), then do any of the following:
 - *Reschedule it*: Choose Edit Time, choose a new time, then click away from the schedule to commit the new time.
 - *Send it immediately*: Choose Send Message.

- **Edit a scheduled message**

You can edit a scheduled message until its delivery time.

1. Go to the Messages app  on your Mac.
2. Select the conversation with the message you want to edit.
If you can't see your message, scroll down to the bottom of the conversation. Scheduled messages may appear below sent messages.
3. Control-click the message you want to change, then choose Edit.
4. Make any changes, then click  to resend with edits or  to revert. See Edit a sent message.

- **Delete a scheduled message**

When you delete a message before its scheduled send time, it's cancelled and not delivered to the recipient.

1. Go to the Messages app  on your Mac.
2. Select the conversation with the message you want to delete.
If you can't see your message, scroll down to the bottom of the conversation.
Scheduled messages may appear below sent messages.
3. Click Edit in the conversation (above your scheduled message), then choose Delete Message.

- **Solve maths in Notes on Mac**

With macOS Sequoia, you can crunch numbers, assign variables and evaluate expressions when you use upgraded iCloud notes or notes stored on your Mac. Just enter a formula or an expression followed by an equals sign while typing to have it instantly solve in your note body. You can also customize how you view your maths results, or turn them off.

Note: Math Results work with math written using Western Arabic numerals and common mathematical symbols, or typed using Arabic (Eastern & Western) and Devanagari numerals and common mathematical symbols.

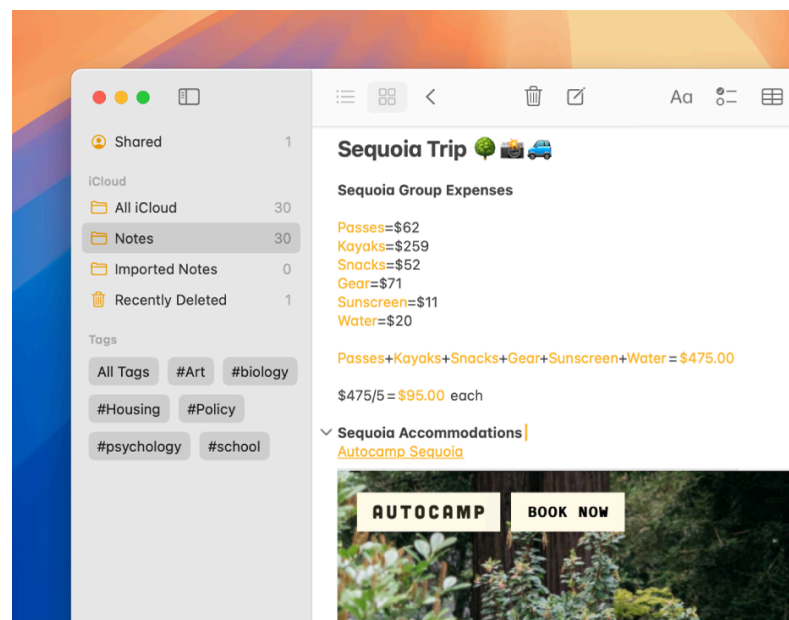


Figure 3.10 solve in notes on Mac, Source: Apple Support (2024), ref at: <https://support.apple.com/en-gb/guide/notes/apda85974595/mac>

- **Solve expressions**

You can solve mathematical expressions with parentheses, and basic and scientific operations.

Note: Notes uses the standard order of operations. For example, exponents are completed before multiplication and division, and multiplication and division are completed before addition and subtraction. Add parentheses around expressions that you want calculated first.

1. Go to the Notes app  on your Mac.
2. Open a note, or click  to create a new note.
3. Type a math problem, followed immediately by the equals sign.
For example, “ $(27/3)^2=$ ” or “ $47 \times 96=$ ”.
4. By default, answers appear as suggestions. Press Return to insert the answer inline.
You can choose to have answers immediately inserted after an equals sign or turn them off. See Choose how Maths Results appear.

- **Define a variable**

Note: Variables are available when created using the Latin alphabet.

1. Go to the Notes app  on your Mac.
2. Open a note, or click  to create a new note.
3. To define a variable, type a letter or a word, followed immediately by the equals sign and the variable’s value.
For example, “Lunch=25” or “x=2.”

Note: For Notes to recognise the variable, the variable must come before (to the left of or above) an expression that uses that variable.

The word or letter changes colour when Notes recognises it as a variable.

Solve expressions with variables

- **Solve complex expressions with variables.**

1. Go to the Notes app  on your Mac.
2. Open a note, or click  to create a new note.
3. Define a variable.
Note: For Notes to recognise the variable, the variable must be defined before (to the left of or above) an expression that uses that variable.
4. Type your expression including the variable, followed immediately by the equals sign.
 For example, “Passes+Kayaks=”
 The answer to your expression appears inline after the equals sign.
5. Press Return to insert the answer.

When you change the value of a variable that precedes the expression, the expression’s answer updates automatically.

Tip: You can quickly change the value of a variable by double-clicking the number, then using the arrows to increase or decrease the variable’s value.

Choose how Maths Results appear

By default, Maths Results appear as suggestions, but you can choose to have Notes immediately insert results or even turn them off.

1. Go to the Notes app  on your Mac.
2. Choose Format > Maths Results, then choose one of the following:
 - *Insert Results:* Immediately inserts results after an equals sign.
 - *Suggest Results:* Suggests results after an equals sign, which can be inserted by pressing Return.
 - *Off:* Turns off Maths Results so that they do not appear. (*Apple Inc. 2024*).

CHAPTER IV

CONCLUSION AND SUGGESTION

4.1 Conclusion

As new students beginning our journey in information technology, working on this ISAS gave us a wider perspective on what an operating system really is. at first, we saw an OS simply as software that makes a computer run, but by learning about macOS Sequoia we realized that it is also a platform that connects devices, supports productivity, and reflects how technology evolves. The new features such as Apple Intelligence, iPhone Mirroring, and stronger security showed us both the strengths of innovation and the limits of compatibility, especially for users with older devices. This experience has taught us that understanding an OS is not only about knowing its features, but also about appreciating the balance between progress and accessibility.

4.2 Suggestion

From what we have studied, we suggest that Apple could make macOS Sequoia more inclusive by extending support to older devices and adding features that help students in learning. For ourselves as beginner students, we believe it is important to keep exploring not only macOS but also other operating systems such as Windows and Linux, so we can build a stronger foundation of knowledge. By staying curious and open to learning, we will be better prepared for the rapid changes in technology in the years ahead.

BIBLIOGRAPHY

Apple Inc. 2025. *Mac User Guide.*

[Mac User Guide](#) [29/08/2025]

Ashabul Taufik et al. 2025. *Education Journal of Education and Technology.*

["Dasar Sistem Operasi", Journal of Education and Technology](#) [06/09/2025]

Wikipedia org. 2024. *MacOS*

[Wikipedia - MacOS](#) [29/08/2025]

Silberschatz, Galvin, & Gagne. 2005. *Operating System Concepts, Seventh Edition.*

[Operating System Concepts, Seventh Edition](#) [29/08/2025]

Blog Routine hub. 2024. *What is Apples WWDC?*

[What is Apple WWDC?](#) [09/09/2025]